

Branch Specialization Checkpoint

Weakly Supervised Branch Routing

Branch Specialization Project

2026-05-22

Checkpoint reason: weak routing supervision produced near-oracle learned modularity.

Previous checkpoint:

- unconstrained learned routers solved the task;
- they did not recover role-specific branch modularity;
- oracle routing showed that modularity is possible in this setup.

This checkpoint asks:

Is a small scored-position routing signal sufficient for learned functional modularity to emerge?

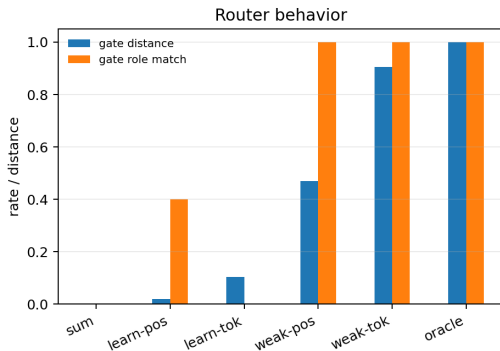
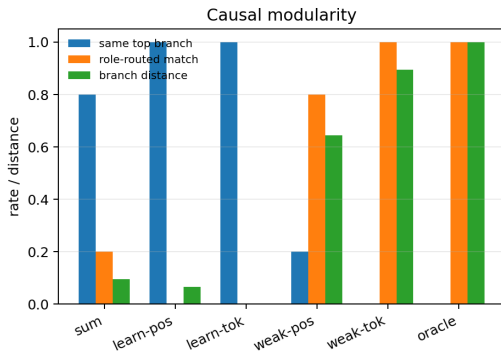
Same local-vs-induction competition task:

- local copy positions predict x from $[x, \text{SEP}, x]$;
- induction positions predict the next token in a repeated random sequence;
- local weight 0.25, induction weight 1.0;
- five seeds per config, 1200 training steps.

New intervention:

- weak auxiliary routing loss, weight 0.05;
- local scored positions target branch 0;
- induction scored positions target branch 1;
- branch modularity is still measured by causal branch ablation.

Main Comparison



Weak token routing moves from the entangled learned-router regime to near the oracle regime on both gate behavior and causal branch modularity.

Modularity Table

Config	Local acc.	Ind. acc.	Same top	Routed match	Branch distance
branch_sum	1.0000	0.9997	0.80	0.20	0.0951
learned_pos	0.9999	0.9987	1.00	0.00	0.0665
learned_tok	1.0000	0.9989	1.00	0.00	0.0006
weak_pos	0.9999	0.9988	0.20	0.80	0.6446
weak_tok	0.9998	0.9983	0.00	1.00	0.8957
oracle	1.0000	0.9988	0.00	1.00	1.0000

Routed match means local depends most on branch 0 and induction depends most on branch 1 under ablation.

Gate Behavior

Config	Gate match	Gate distance	Local entropy	Ind. entropy	Target NLL
weak_pos	1.00	0.4700	0.4811	0.6422	0.3139
weak_tok	1.00	0.9049	0.0125	0.1944	0.0634

The token router learned much sharper gates. The position router followed the target direction, but remained softer and failed in one of five seeds.

Branch Ablation

Config	Local B0	Local B1	Induction B0	Induction B1
learned_tok	2.2221	2.7767	2.3307	3.2506
weak_pos	0.9813	0.0406	0.5780	1.3542
weak_tok	5.6793	0.0000	0.5184	4.8444
oracle	5.8198	0.0000	0.0000	6.1639

Weak token routing makes local branch 0 almost oracle-like. Induction mostly uses branch 1, but still has residual dependence on branch 0.

Supported in this toy setup:

- Weak scored-position routing supervision is sufficient for learned branch-level functional modularity.
- Token-dependent routing is cleaner than position-only routing.
- Gate modularity and causal modularity now move together for the token router.

Not shown:

Spontaneous modularity without role labels. The auxiliary loss directly names the desired role split.

Continue Phase 3 with two sharper tests:

- ① sweep lower supervision weights to estimate the minimum signal needed;
- ② test unlabeled routing pressure, such as entropy or load-balancing regularization, and a more conflict-heavy task.

Current working claim:

Learned functional modularity is possible here, but it needs routing pressure; gates without pressure did not find it on their own.

Sources and Provenance

- Report: `doc/phase3_toy_weak_router.md`
- Model script: `scripts/toy_branch_isolation_intervention.py`
- Weak-router output: `results/phase3_toy_weak_router_w005_lw025/`
- Combined comparison: `results/phase3_toy_weak_router_w005_combined/`
- Deck data snapshot: `presentations/2026-05-22-1300-weak-router/data/`